CURRICULUM VITAE

LEI GIOIA YANG

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Websites	Google Scholar , arXiv, ORCID, Inspire-Hep
Citizenship	New Zealand

Research interests

Theoretical quantum matter; quantum information; mathematical physics; quantum phases of matter; strongly interacting many-body systems; properties of many-body Hamiltonians and states; entanglement measures; quantum anomalies; quantum many-body scars; topological semimetals; chiral fermions; crystalline defects; quantum circuit lower bounds; quantum state preparation; quantum simulation.

Education and work

2026 - current	**Postdoctoral Fellow**, *University of Boulder Colorado*, USA.
2023 - 2026	**Burke Postdoctoral Fellow**, Sherman Fairchild Postdoctoral Scholar Research Associate in Theoretical Physics (100% Research), *California Institute of Technology*, USA.
2023 (10th October)	**PhD in physics**, *University of Waterloo & Perimeter Institute*, Canada.
2018	**MSc in physics,** Perimeter Scholars International program, *University of Waterloo & Perimeter Institute*, Canada.
2017	**MSc in physics**, *Victoria University of Wellington*, New Zealand.
2015	**BSc (Hons) in mathematics and physics**, *Victoria University of Wellington*, New Zealand.

Upcoming appointment

2027 - onwards	**Assistant Professor**, Department of Physics and Astronomy, *Tufts University*, Boston USA.

Awards and competitive funding

2023	Burke Postdoctoral Fellowship, *California Institute of Technology*, USA.
2023	KITP Graduate Fellowship, *Kavli Institute for Theoretical Physics*, USA.
2019	Vanier Canada Graduate Scholarship, *Natural Sciences and Engineering Research Council of Canada (NSERC)*, Canada, National institutionally nominated doctoral scholarship; CAD 150,000.
2019	President's Graduate Scholarship, *University of Waterloo*, Canada; CAD 30,000.
2017	Perimeter Scholars International, *Perimeter Institute/University of Waterloo*, Canada Fully-funded MSc scholarship; CAD 45,000.

Published or accepted papers

Note: In reverse-chronological order. I publish under the name Lei Gioia.

- **Lei Gioia**, Anton A. Burkov, and Taylor L. Hughes, *No-go theorem for single time-reversal invariant symmetry-protected Dirac fermions in 3+1d*, accepted for publication in *Physical Review B* (2026), DOI: 10.1103/g4h4-kb3c
- **Lei Gioia** and Ryan Thorngren, *Exact Chiral Symmetries of 3+1D Hamiltonian Lattice Fermions*, Phys. Rev. Lett. **136** (2026), DOI: 10.1103/s5q2-317m
- Julian May-Mann, Mark R. Hirsbrunner, **Lei Gioia**, Taylor L. Hughes, *Crystalline axion electrodynamics in charge-ordered Dirac semimetals*, Phys. Rev. B **110** (2024), DOI: 10.1103/PhysRevB.110.155110
- Jinmin Yi, Xuzhe Ying, **Lei Gioia**, and A. A. Burkov, *Topological order in interacting semimetals*, Phys. Rev. B **107** (2023), DOI: 10.1103/PhysRevB.107.115147
- **Lei Gioia** and Chong Wang, *Non-zero momentum requires long-range entanglement*, Phys. Rev. X **12** (2022), DOI: 10.1103/PhysRevX.12.031007
- **Lei Gioia**, Chong Wang, and A. A. Burkov, *Unquantized anomalies in topological semimetals*, Phys. Rev. Research **3** (2021), DOI: 10.1103/PhysRevResearch.3.043067
- Chong Wang, **Lei Gioia**, and A. A. Burkov, *Fractional Quantum Hall Effect in Weyl Semimetals*, Phys. Rev. Lett. **124** (2020), DOI: 10.1103/PhysRevLett.124.096603
- **Lei Gioia**, M. G. Christie, U. Zülicke, M. Governale, and A. J. Sneyd, *Spherical topological insulator nanoparticles: Quantum size effects and optical transitions,* Phys. Rev. B **100** (2019), DOI: 10.1103/PhysRevB.100.205417
- **Lei Gioia**, U. Zülicke, M. Governale, and R. Winkler, *Dirac electrons in quantum rings*, Phys. Rev. B **97** (2018), DOI: 10.1103/PhysRevB.97.205421

Preprints

- Ryan Thorngren, **Lei Gioia**, and Carolyn Zhang, *Translation symmetry-enforced long-range entanglement in mixed states*, arXiv:2605.15200 (2026) (in PRL review)*
- Nicholas O'Dea, **Lei Gioia**, Sanjay Moudgalya, and Olexei I. Motrunich, *Locality forces equal energy spacing of quantum many-body scar towers*, arXiv:2601.14206 (2026) (in PRX Quantum review)*
- **Lei Gioia**, Sanjay Moudgalya, and Olexei I Motrunich, *Distinct Types of Parent Hamiltonians for Quantum States: Insights from the W State as a Quantum Many-Body Scar*, arXiv:2510.24713 (2025) (in PRB review)*
- **Lei Gioia** and Ryan Thorngren, *W State is not the unique ground state of any local Hamiltonian*, arXiv:2310.10716 (2023) (in PRL review)*

Dissertations

2023	**PhD thesis**, L. Yang, *Interactions, Entanglement, and Anomalies in Topological Semimetals*, University of Waterloo, URL.
2018	**MSc thesis**, L. Yang, *Quantum anomalies in Type I Dirac semimetals*, University of Waterloo.
2017	**MSc thesis**, L. Yang, *Quantum confinement of chiral charge carriers in ring conductors,* Victoria University of Wellington, URL.

Selected Presentations

2026	**Workshop talk**, "Exact Chiral Symmetries of 3+1D Hamiltonian Lattice Fermions", YITP: Frontiers of Lattice Fermions, *Kyoto University, Japan*
2026	**Physics colloquium**, "Renormalization, symmetries, and anomalies: Organizing principles for entangled quantum matter", *Tufts University, Boston*
2026	**Physics colloquium**, "A window into quantum matter: the renormalization group view", *New York University, New York*
2025	**Blackboard talk**, "Exact chiral symmetries on the lattice", Lattice QCD talk, *Massachusetts Institute of Technology, Boston*
2025	**Condensed matter seminar**, "Distinct Types of Parent Hamiltonians for Quantum States", *Boston University, Boston*
2025	**Talk**, "W State as a many-body quantum scar", IQIM annual retreat, *IQIM & Caltech, Lake Arrowhead*
2025	**Workshop talk**, "The RG map: a window into gapless states", Simons Foundation: Ultra Quantum Matter meeting, *University of Chicago, Chicago*
2024	**Workshop talk**, "W-state is not the unique ground state of any local Hamiltonian ", C*-Algebraic Quantum Mechanics and Topological Phases of Matter workshop, *University of Colorado Boulder*, *Colorado*
2023	**Quantum information seminar**, "W-state is not the unique ground state of any local Hamiltonian", *Max-Planck of Quantum Optics*, *Munich*
2023	**Condensed matter seminar**, "Momentum creates long-range entanglement and quantum anomalies", *University of California Berkeley, Berkeley*
2023	**Condensed matter seminar**, "Momentum creates long-range entanglement and quantum anomalies", *University of California Santa Cruz, Santa Cruz*
2022	**Condensed matter seminar**, "Momentum creates long-range entanglement and quantum anomalies", *University of Zurich*, *Zurich*
2022	**Blackboard seminar**, "Non-zero momentum requires long-range entanglement", *Kavli Institute for Theoretical Physics, Santa Barbara*

2022 **Condensed matter seminar**, “Momentum creates long-range entanglement and quantum anomalies”, *California Institute of Technology, Pasadena*

2022 **Condensed matter seminar**, Institute for Condensed Matter Theory, “Momentum creates long-range entanglement and quantum anomalies”, *University of Illinois Urbana-Champaign*

Teaching and supervision

2022 **Research project supervisor**, PSI Winter School, *Machine learning social dynamics in physics*, One week supervision of two students in understanding the dynamics of physicists, their research, and their publishing behaviors.

2012 – 2017 **Laboratory and tutorial demonstrator; outreach demonstrator**, *School of Chemical and Physical Sciences*,
Victoria University of Wellington, NZ
Demonstrated and tutored for courses:

- Phys 114 - Physics 1A
- Phys 115 - Physics 1B
- Phys 221 - Relativity and Quantum Physics
- Phys 223 - Classical Physics
- Phys 309 - Solid State & Nuclear Physics

Academic services

2025 - current **Committee member**, IQIM seminar series, Helped administer a USD 40,000 speaker budget and host researchers,
Institute of Quantum Information and Matter (IQIM) & Caltech.

2020 - 2021 **Committee member**, Internal Strategic Planning Committee, *Perimeter Institute for Theoretical Physics*.

2019 - 2021 **Committee chair**, Ph.D. student committee, Elected, Represented graduate students and helped establish a peer-mentoring programme, *Perimeter Institute for Theoretical Physics*.

2019 - 2021 **Committee member**, Respectful Environment working group, Co-authored the Respectful Environment code of conduct, *Perimeter Institute for Theoretical Physics*.

2019 **Event organizer**, FemPhys mentoring event, organised an event with 20 female mentors & 50 mentees, *Perimeter Institute for Theoretical Physics*.